- **Clean up.** When the exercise is finished, delete the test files and the role assignment, or delete the resource group, so nothing keeps running or stays accessible.

SESSION 3 — AZURE II: SECURITY AND DATA INTEGRATION

- **Topic:** Azure II — Security and Data Integration
- **Date:** 23rd October 2026

10. Azure Key Vault Fundamentals: Secrets and Keys

In simple words. Azure Key Vault is a locked safe in the cloud for sensitive values. Instead of writing a password or a key inside your code, a configuration file, or a pipeline, you store it once in the vault, and the application asks the vault for it when it needs it. Access is checked every time, every request is logged, and the value can be changed in one place without touching the applications that use it.

Object	**Simple meaning**
Secret	A small piece of sensitive text that you want to read back later, such as a password, a connection string, or an API key. Applications fetch the value from the vault when they run.
Key	A cryptographic key used to lock (encrypt) and unlock (decrypt) data, or to sign it. For a key, you do not take the key out of the vault. You send data to the vault and ask it to do the encrypting or signing for you, so the key itself stays protected.
Certificate	A digital certificate, for example one that proves a website's identity, stored with its renewal details so that it can be renewed before it expires.
Secrets vs. keys	**The key difference**
Secret	You retrieve the value and use it yourself. Example: an application reads a database password from the vault and uses it to connect.
Key	You ask the vault to use the key for you. Example: a storage account asks the vault to unlock the key that encrypts its data, which is called a customer-managed key.

Versions and identifiers. Every time a secret or key is changed, the vault keeps the old value and creates a new version, so nothing is lost by accident and a rotation can be rolled back. Each item has an address of the form **https://<vault-name>.vault.azure.net/secrets/<secret-name>**.

A secret can also have an activation date and an expiry date, so that it stops working on a planned day.

Protection against deleting by mistake. Two safety features protect a vault. **Soft delete means a deleted vault or item is kept for a retention period, 90 days by default, and can be recovered**. It is always on for new vaults and cannot be turned off. **Purge protection is an optional extra: when on, nobody can permanently erase a deleted vault or item until the retention period ends**, and once enabled it cannot

be turned off. Both are especially important if encryption keys are stored in the vault, because losing the key would make the encrypted data unreadable.

- **One vault per environment.** Use separate vaults for development and production, so a mistake or a leaked development credential cannot reach production secrets.
- **Rotate and expire.** Set expiry dates and replace secrets on a schedule. Because applications read the latest version, a rotation needs no code change.
- **Limit the network.** A vault can be locked to chosen networks or reached only through a private endpoint, so that it is not open to the whole internet.

11. Key Vault Access: RBAC vs. Access Policies

Both models answer the same question: who may do what with the items in this vault. They are two different ways of recording the answer, and a vault uses only one of them at a time. Azure RBAC is the same role system from starting, now applied to a vault. Access policies are the older, vault-only system.

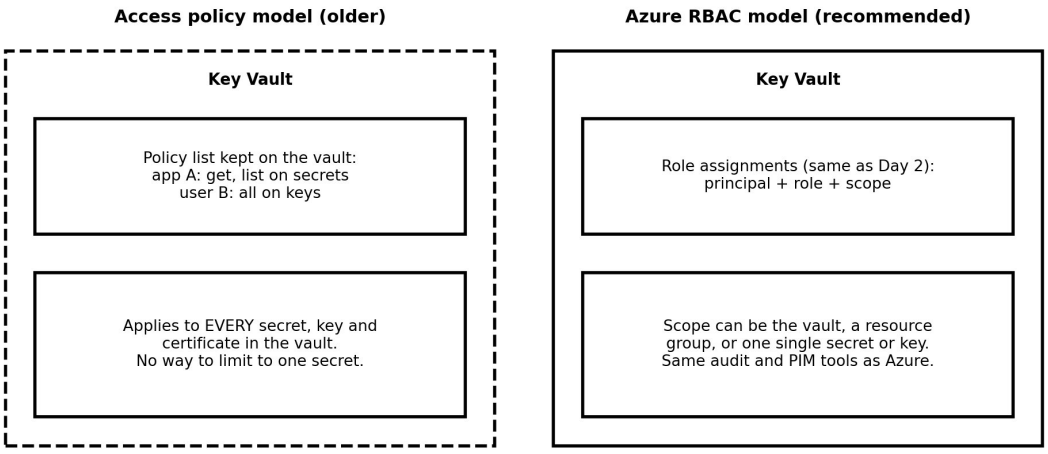


Diagram 8: Access policies apply to everything in the vault, while RBAC roles can be scoped as narrowly as one secret.

Point	**Access policies vs. Azure RBAC**
Where it is set	Policies: in a list stored on the vault itself. RBAC: as ordinary role assignments, in the same Access control (IAM) place as the rest of Azure.
How fine the control is	Policies: one entry covers the whole vault, so an identity that may read one secret can read them all. RBAC: a role can be given at the vault, the resource group, or a single secret or key.
What an entry contains	Policies: a list of permissions such as **get, list, and set** for keys, secrets, and certificates.

	RBAC: a named role such as Key Vault Secrets User.
Who can change it	Policies: managed through the management side of Azure, so someone with a broad management role such as Contributor can give themselves data access. RBAC: changing access needs a role-assignment permission such as Owner or User Access Administrator, which is a stronger separation.

Management side and data side. Azure separates managing the vault from using what is inside it. Creating the vault, changing its settings, and deleting it always go through Azure RBAC. Reading secrets and using keys, the data side, can use either model. This is why someone who is Owner of a vault can still be refused when they try to read a secret in an RBAC vault.

Key Vault role	**What it allows**
Key Vault Administrator	All data operations on secrets, keys, and certificates.
Key Vault Secrets Officer	Create, change, and delete secrets, but not manage who has access.
Key Vault Secrets User	Read the value of secrets. This is the role an application normally needs.
Key Vault Crypto Officer	Create and manage keys, and use them.
Key Vault Crypto User	Use keys for encrypting, decrypting, and signing, without managing them.
Key Vault Certificates Officer	Manage certificates.
Key Vault Reader	See that items exist and read their details, but not read secret values or use keys.

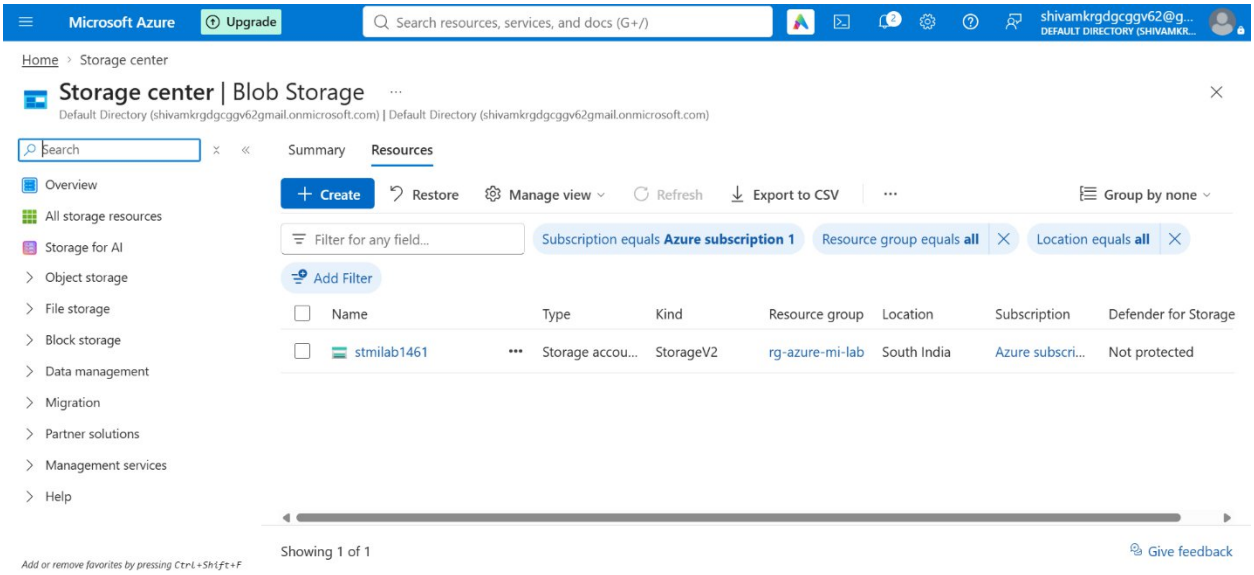
- **Switching has a cost.** Turning on the RBAC model makes every existing access policy stop working. Assign the equivalent roles first, then switch, or applications will suddenly lose access.
- **Map policies to roles.** Read-only policies on secrets become Key Vault Secrets User, secret management becomes Secrets Officer, and key operations become Crypto User or Crypto Officer.

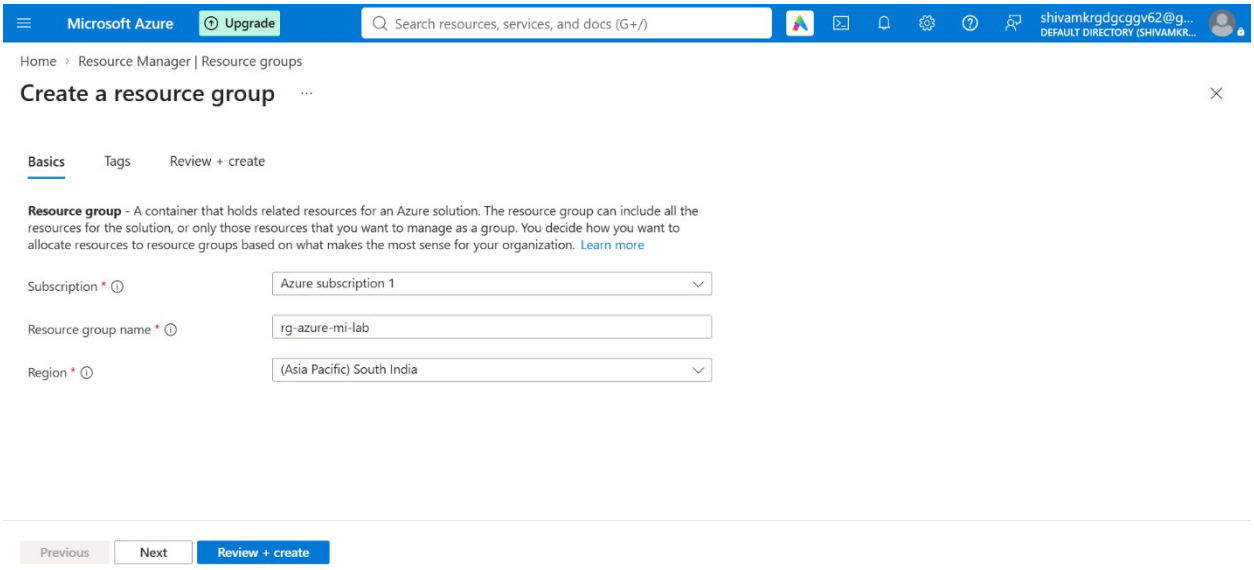
Which to choose. Use RBAC for new vaults. Access policies are mainly something you will meet in older projects that have not been migrated yet.

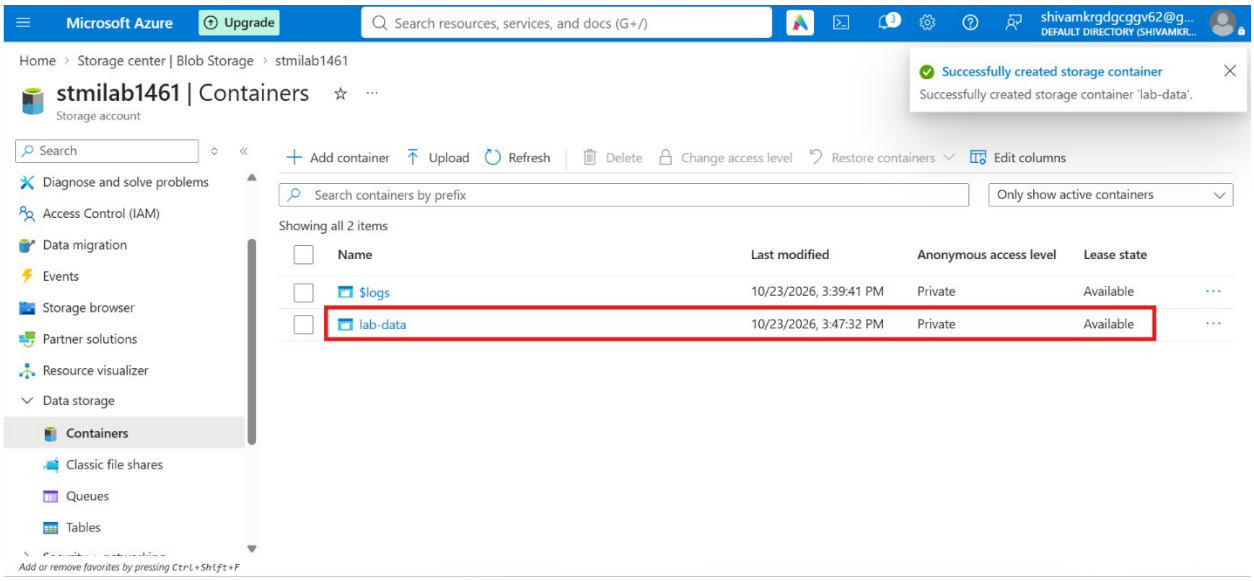
Lab 1 (Azure) overview

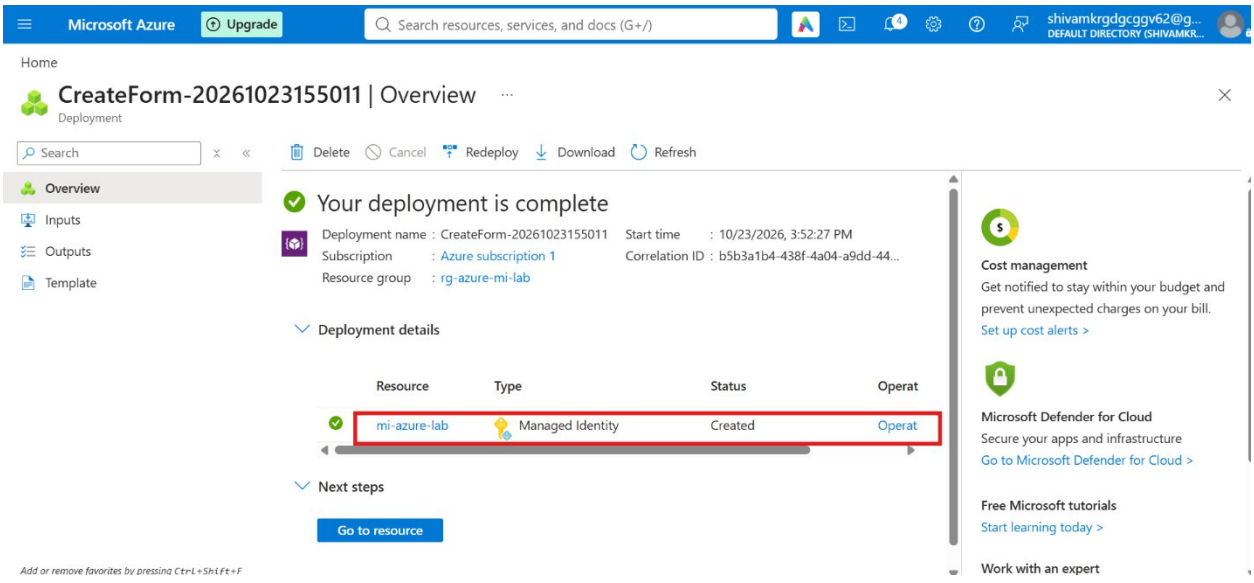
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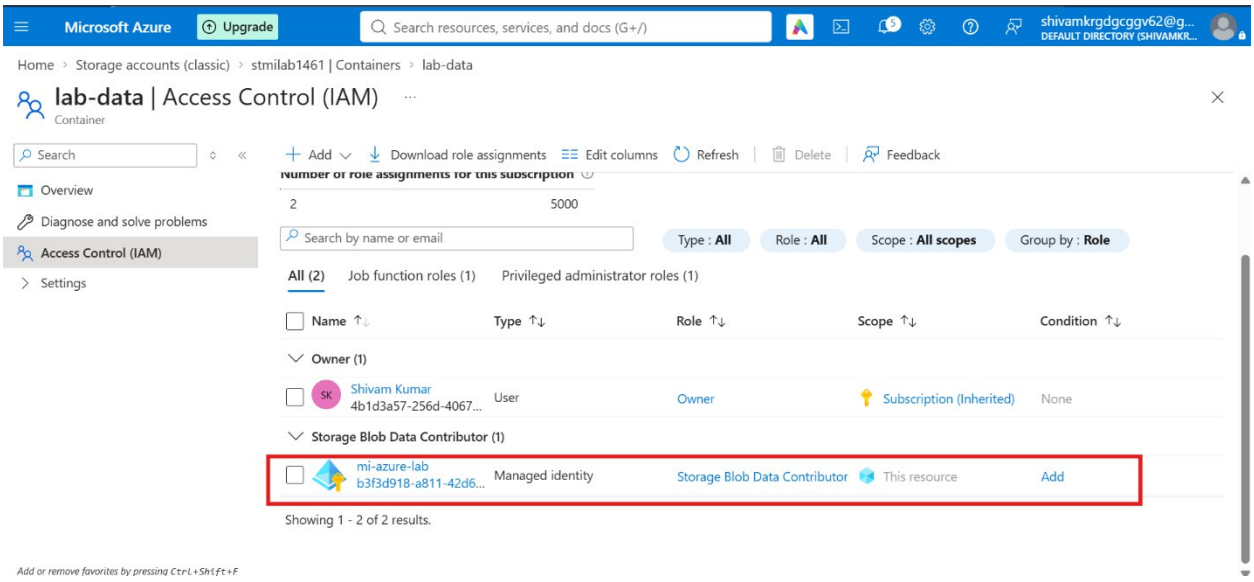
Create an Azure Storage Account with ADLS Gen2 enabled, create a User-Assigned Managed Identity (UAMI), assign the Storage Blob Data Contributor role at the container scope, and verify access through a Python script.

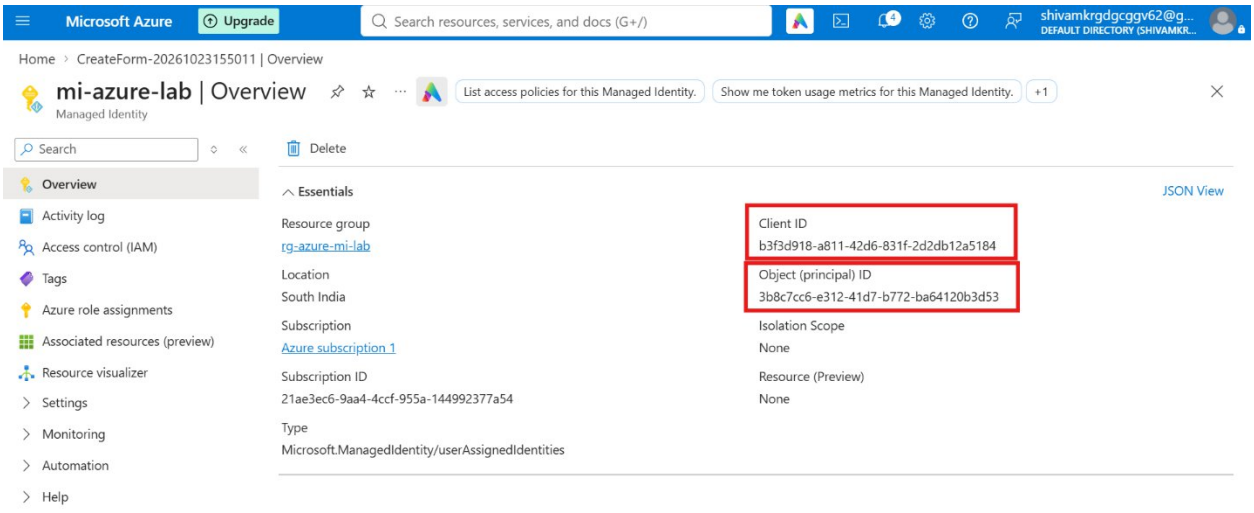


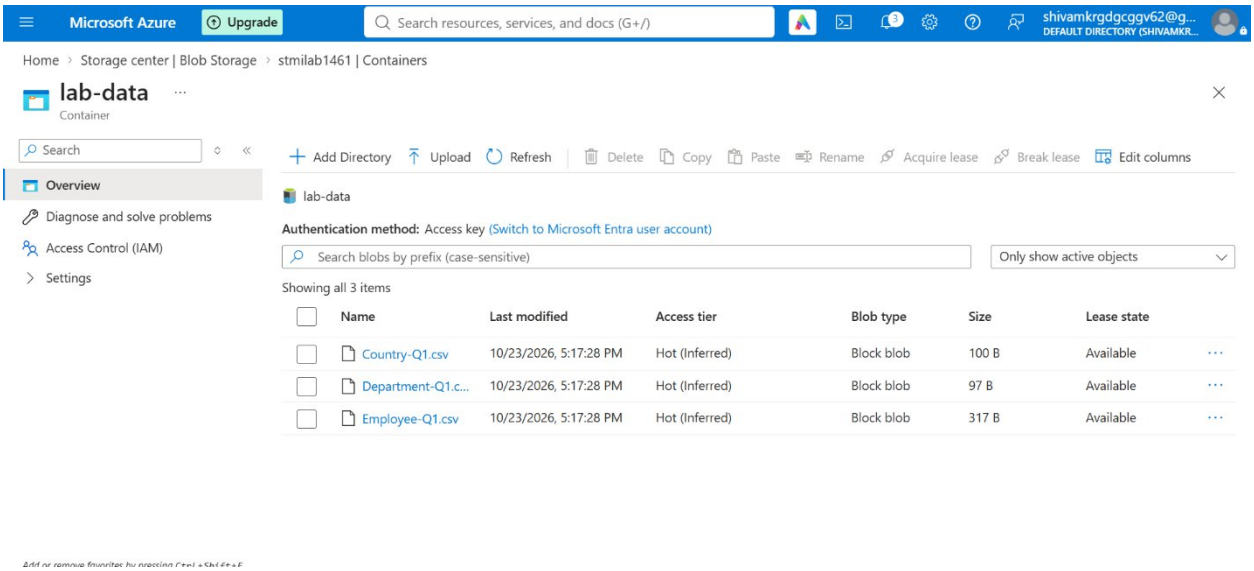



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